

L1 List-Side Compilers and Sunflower Residuals

RS-MCA experimental note
integrated from AllenGrahamHart L1 PRs, with Codex editing

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Abstract

This note consolidates the L1/list-side material that may be useful for a future CAP25 revision. The proved parts are compiler statements: quotient-core planted lower counts, exact 2^{-128} budget windows, ordinary Johnson-region upper bounds, the sunflower auxiliary-list reduction, the few-petal Johnson regime, and fixed cofactor-excess full-petal layers. The experimental parts stress-test the remaining mixed-petal and growing-defect sunflower residuals.

The message for CAP25 is conservative. The list-side planted obstruction gives clean unsafe rows and exact arithmetic windows. The safe side still needs the full image-fiber theorem for primitive, non-planted lists. Petal/PMA packets show which sunflower layers are already controlled and which layers remain as the true L1 obstruction.

1 Status and Role

Let $C = \text{RS}[F, H, k]$ be a Reed–Solomon code of length $n = |H|$ and let $s = k + \sigma$ be an agreement threshold. For a received word $U : H \rightarrow F$, write

$$\text{ImgFib}_U(s) = \{P \in F[X]_{\deg < k} : |\{x \in H : U(x) = P(x)\}| \geq s\}.$$

This is the actual list object; it avoids raw support multiplicity.

The L1 positive target is not that quotient-periodic lists are small. It is that, after quotient-periodic/stabilized structure is paid, the primitive remainder is polynomial once the generated-field reserve and lower cutoff clear.

Remark 1.1 (CAP25 fit). In Paper D/CAP25, L1 is a supporting safe-side and list-track input. The material here can supply lower-side planted rows and bounded residual compiler pieces, but it does not prove the final safe-side image-fiber theorem.

2 Planted Quotient-Core Arithmetic

Theorem 2.1 (planted quotient-core lower count). *Let $C = \text{RS}[F, H, k]$, $|H| = n$, and suppose $M \mid \gcd(n, k)$ satisfies*

$$1 \leq \sigma < M, \quad k/M \leq n/M - 1.$$

Then there is a received word with at least

$$P_M(n, k) = \binom{n/M - 1}{k/M}$$

codewords agreeing on $k + \sigma$ positions. Consequently

$$\max_U |\text{ImgFib}_U(k + \sigma)| \geq \max_{\substack{M \mid \gcd(n, k) \\ \sigma < M \\ k/M \leq n/M - 1}} \binom{n/M - 1}{k/M}.$$

Proof sketch. Choose a subgroup $K \leq H$ of order M , a distinguished K -coset C_0 , and a set $T \subset C_0$ of size σ . For every k/M -subset of the quotient $H/K \setminus \{C_0\}$, the quotient-core construction gives a distinct degree- $< k$ codeword agreeing with the planted word on $k + \sigma$ positions. This gives the stated binomial count. $\square$

Corollary 2.2 (list unsafe test). *For a list denominator q_{list} and target 2^{-128} , the planted quotient-core packet proves a list row unsafe whenever*

$$\max_M \binom{n/M - 1}{k/M} > \left\lfloor \frac{q_{\text{list}}}{2^{128}} \right\rfloor.$$

Proposition 2.3 (dyadic planted crossings). *For $n = 2^m$, $k = \rho n$, and*

$$\rho \in \{1/2, 1/4, 1/8, 1/16\},$$

the first dyadic quotient orders at which the planted count beats the largest budget $2^{128} - 1$ are

ρ	1/2	1/4	1/8	1/16
$N = n/M$	256	256	256	512.

Remark 2.4. These are lower-side certificates only. Above the displayed boundary, the planted dyadic scale may deactivate, but safety requires the primitive image-fiber theorem and the non-planted extras ledger.

3 Budget Windows

If a planted exact count is P and the certified lower count is $L \leq P$, then the only unresolved list budgets are

$$L \leq \lfloor q_{\text{list}}/2^{128} \rfloor < P.$$

Equivalently,

$$L2^{128} \leq q_{\text{list}} \leq P2^{128} - 1.$$

This is the list-side analogue of the threshold compiler in ‘thresholds.tex’: no finite-field value-set issue remains once the planted word is constructed.

4 Ordinary Johnson Region

Theorem 4.1 (full-list Johnson region). *Let H be any Reed–Solomon evaluation domain of size n and $C = \text{RS}[H, k]$. For every received word U :*

(i) *if $2s > n + k - 1$, then $|\text{ImgFib}_U(s)| \leq 1$;*

(ii) *if $s^2 > n(k - 1)$, then*

$$|\text{ImgFib}_U(s)| \leq \frac{n(n - k + 1)}{s^2 - n(k - 1)}.$$

Proof. For listed polynomials P_i , let A_i be their agreement sets with U . Then $|A_i| \geq s$ and $|A_i \cap A_j| \leq k - 1$ for $i \neq j$. The unique decoding claim follows from $2s - n > k - 1$. For the Johnson bound, write $m_x = \#\{i : x \in A_i\}$ and $I = \sum_x m_x$. Then $I \geq Ls$ and

$$\sum_x \binom{m_x}{2} \leq \binom{L}{2} (k - 1).$$

Cauchy-Schwarz gives

$$L^2 s^2 \leq n \sum_x m_x^2 \leq n(Ln + L(L - 1)(k - 1)),$$

which rearranges to the displayed bound. $\square$

Remark 4.2. The L1 problem begins below the ordinary Johnson region. Above it, the primitive list remainder is already polynomial by Theorem 4.1.

5 Sunflower and Petal Reductions

The remaining L1 program analyzes non-planted list words through sunflower decompositions. The following compiler statements are the pieces that can be used now.

5.1 Auxiliary petal-list reduction

Fix a missed-core set D of size d and a retained residual subset R_0 of size r . Let L_D be the missed-core locator. Suppose petals $T_1, \dots, T_M$ have size $\sigma + 1$ and scalar labels c_i . Define the auxiliary word on $T = \bigcup_i T_i$ by

$$U_D(x) = c_i L_D(x), \quad x \in T_i.$$

Proposition 5.1 (sunflower core-defect reduction). *Every non-planted listed codeword in the fixed (D, R_0) layer injects into an auxiliary Reed-Solomon list of degree at most d on the petal domain T , at agreement*

$$a = \sigma + d + 1 - r.$$

Dropping the extra zero constraints on R_0 only enlarges this auxiliary list, so it is safe for upper bounds.

5.2 Few-petal Johnson region

The auxiliary petal domain has size $|T| = M(\sigma + 1)$ and effective dimension $d + 1$.

Corollary 5.2 (few-petal Johnson boundary). *If*

$$a^2 > |T|d,$$

then the fixed (D, R_0) auxiliary list has size at most

$$\frac{|T|(|T| - d)}{a^2 - |T|d}.$$

Equivalently, for $d > 0$, the few-petal Johnson-covered range is

$$M \leq \left\lfloor \frac{a^2 - 1}{d(\sigma + 1)} \right\rfloor.$$

Proof. Apply Theorem 4.1 to the auxiliary degree- $\leq d$ code, whose dimension is $d + 1$, and use the strict integer form of the Johnson inequality. $\square$

5.3 Fixed cofactor-excess full-petal layers

Let $d = \ell + e$, where ℓ is the petal size and e is the cofactor excess.

Proposition 5.3 (fixed-excess full-petal compiler). *For every fixed $E \geq 0$, the total number of non-planted listed codewords whose touched petals are all full and whose defect satisfies $\ell \leq d \leq \ell + E$ is at most*

$$\binom{M}{2} q + 2^M \sum_{e=1}^E q^{e+1}.$$

Equivalently, the layer bounds are

$$e = 0 : \binom{M}{2} q, \quad e \geq 1 : 2^M q^{e+1}.$$

Corollary 5.4. *At the intended L1 lower cutoff, $M = O(\log n)$. If $q \leq n^Q$ and E is fixed, then the fixed-excess full-petal contribution is polynomial in n . Thus a full-petal counterexample must force $e = d - \ell$ to grow with n .*

6 Evidence and Route Cuts

6.1 Wide-regime auxiliary evidence

The petal auxiliary wide-regime scan over F_{109} tests whether auxiliary lists behind the sunflower residual become large after the ordinary Johnson cutoff. The tested rows show:

- (i) Johnson-covered rows are empty;
- (ii) wide rows are populated;
- (iii) in the exact window, the largest list size is $2023 < 36^3$.

This is evidence for a controlled mixed-petal residual, not a proof of polynomial growth.

6.2 Worst-word challenge

The small F_{193} sweep refutes the literal statement that the planted word is always worst: at $n = 16$, $\sigma = 1$, several cells beat the planted count. The exceptions are mixed-petal sunflower extras. For $\sigma \geq 2$ in the tested exact cells, no challenger beats the planted count, and larger structured cells at $n = 32, 64$ found no non-planted structured challenger.

Remark 6.1 (corrected lesson). The safe formulation is not “planted is always worst.” It is that low-slack mixed-petal extras are a named residual branch, while planted quotient cores remain the clean lower-side certificate.

6.3 Growing-defect full-petal witnesses

Explicit full-petal witnesses exist with positive cofactor excess $d - \ell > 0$: one at $d - \ell = 2$ with three petals, and one at $d - \ell = 5$ with five petals. They show the full-petal growing-defect branch is not vacuous. They do not establish an asymptotic growth rate.

7 CAP25 Integration Checklist

A CAP25 revision can use this note as follows:

- (1) Use Theorem 2.1, Corollary 2.2, and Proposition 2.3 for lower-side list caps from planted quotient-core rows.
- (2) Use the budget-window arithmetic to print exact unresolved field intervals for list denominators.
- (3) Use Theorem 4.1 to close ordinary Johnson-region list tails without quotient machinery.
- (4) Use Propositions 5.1 and 5.3 and Corollary 5.2 as paid compiler pieces inside the sunflower residual proof program.
- (5) State the mixed-petal and growing-defect layers as the remaining L1 obstruction; do not hide them inside “planted worst” language.

8 Open Problems

Problem 8.1 (primitive image-fiber theorem). Prove a polynomial bound for the stabilizer-primitive part of $\text{ImgFib}_U(k + \sigma)$ once generated-field entropy reserve and the lower cutoff clear, with quotient-periodic codewords charged separately.

Problem 8.2 (mixed-petal amplification). Classify or bound mixed-petal sunflower extras in the sub-Johnson region, after the few-petal Johnson and fixed-excess full-petal layers are paid.

Problem 8.3 (growing-defect full-petal branch). Determine whether the full-petal branch with $d - \ell \rightarrow \infty$ remains polynomial or produces a new obstruction ledger.

References

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